

AGNS: Empirical Acceleration of Gradient-Regularised Newton Methods with Approximate Hessians

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Abstract—The Gradient-Normalised Smoothness (GNS) framework of [1] provides a single Newton-type update that attains state-of-the-art rates across a broad range of smoothness classes, even when the Hessian is replaced by a cheap PSD approximation; the authors leave acceleration of GNS as future work. We close this gap empirically by introducing AGNS, a heuristic Nesterov-style extension of GNS with O’Donoghue–Candès gradient restart and an optional Sherman–Morrison–Woodbury (WSM) solver for rank-1 Fisher approximations. Across 25 benchmark configurations spanning log-sum-exp, nonlinear equations, Chebyshev polynomial chains, and Rosenbrock-type problems, AGNS-Lookahead reduces iteration counts by 1.3–2.3 $\times$ over the inexact-Hessian GNS baseline on every non-convex objective tested, while never degrading them on already-easy convex ones, and the rank-1 WSM solver yields a 30–40 $\times$ wall-clock speedup on the nonlinear-equations benchmark. The full pipeline is reproducible from a single shell script.

I. INTRODUCTION

Newton-type methods accelerate convergence by exploiting curvature, but the textbook iteration $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k)$ is locally quadratic at best and globally divergent at worst. Three lines of work make Newton’s method globally well-behaved: trust regions [2], cubic regularisation [3], [2], and gradient regularisation [4], [5], [6], [7]. The most recent of these—the Gradient-Normalised-Smoothness (GNS) framework of [1]—unifies the gradient-regularised view across problem classes (Hölder Hessian/third derivative, quasi- and generalised self-concordance, (L_0, L_1) -smoothness), and critically allows the Hessian to be replaced by any PSD approximation $H(x_k) \succeq 0$ without breaking the global rate.

Acceleration is missing. Although Nesterov acceleration [8] is now standard for first-order methods [9], and Monteiro–Svaiter-style acceleration is well-developed for cubic Newton [10], no accelerated variant of GNS exists in the literature: the authors of [1] list it as future work (Section 7).

a) Contribution: We introduce AGNS, a heuristic accelerated variant of GNS obtained by interleaving Nesterov-style extrapolation with the GNS Newton step, anchored at the lookahead point and restarted via the gradient criterion of [11]. Our contribution is empirical, not theoretical: we do not prove a rate for AGNS. Instead, through a fully reproducible pipeline, we demonstrate that AGNS delivers consistent 1.3–2.3 $\times$ iteration speed-ups over inexact-Hessian GNS on every non-convex benchmark we tested, and that combining AGNS

Algorithm 1 AGNS: Accelerated GNS with gradient restart

Require: $x_0 \in \mathbb{R}^n$, $\gamma_0 > 0$, anchor $\in \{\text{lookahead}, \text{iterate}\}$

```

1:  $x_{-1} \leftarrow x_0$ ,  $\ell \leftarrow 0$ 
2: for  $k = 0, 1, \dots$  do
3:    $\beta_k \leftarrow \ell / (\ell + 3)$ ;  $y_k \leftarrow x_k + \beta_k(x_k - x_{k-1})$ 
4:    $z \leftarrow y_k$  if anchor=lookahead else  $x_k$ ;  $H_k \leftarrow H(z)$ 
5:   adaptive search: pick  $\gamma_k$  ensuring (2) at  $y_k$ 
6:    $\lambda_k \leftarrow \|\nabla f(y_k)\|_* / \gamma_k$ 
7:    $x_{k+1} \leftarrow y_k - (H_k + \lambda_k B)^{-1} \nabla f(y_k)$ 
8:   if  $\langle \nabla f(x_{k+1}), x_{k+1} - x_k \rangle > 0$  then
9:      $x_k \leftarrow x_{k+1}$ ;  $\ell \leftarrow 0$  ▷ gradient restart
10:  else  $\ell \leftarrow \ell + 1$ 
```

with a Sherman–Morrison rank-1 solver [12] yields wall-clock speed-ups of 30–40 $\times$ on the nonlinear-equations benchmark.

II. BACKGROUND AND METHOD

For a smooth (possibly non-convex) objective $f: \mathbb{R}^n \rightarrow \mathbb{R}$, metric $B \succ 0$, and PSD Hessian approximation $H(x) \succeq 0$, the GNS update [1] is

$$x_{k+1} = x_k - \left(H(x_k) + \frac{\|\nabla f(x_k)\|_*}{\gamma_k} B \right)^{-1} \nabla f(x_k), \quad (1)$$

with step size $\gamma_k > 0$ chosen by an adaptive search enforcing the *Lemma 1* progress condition

$$f(x_k) - f(x_{k+1}) \geq \frac{\gamma_k}{8} \|\nabla f(x_{k+1})\|_*^2 / \|\nabla f(x_k)\|_*. \quad (2)$$

Practically relevant approximations include the Weighted Gauss–Newton matrix for log-sum-exp [13] and the Fisher approximation for nonlinear-equations problems $f(x) = \frac{1}{p} \|u(x)\|^p$ [14]; both satisfy $\|\nabla^2 f - H\| \leq C_2 \|\nabla f\|_*$, sufficient by [1] to recover the full-Newton rate.

A. The AGNS update

AGNS keeps the GNS adaptive search of (2) unchanged but inserts a Nesterov extrapolation step before the Newton solve. Algorithm 1 gives the full pseudocode.

Hessian anchor. The *lookahead* variant evaluates H at y_k , yielding an exact instance of (1) at y_k and so preserving the Lemma-1 guarantee with respect to $f(y_k)$. The *iterate* variant evaluates H at x_k and is included as an ablation. Empirically (Section III) the lookahead anchor dominates.

Gradient restart. The criterion of [11] resets the local momentum counter ℓ when the new gradient correlates positively with the full step. After a restart $\beta = 0$ on the next iteration, so AGNS degenerates locally to GNS.

Rank-1 WSM solver. For nonlinear equations, $H(x) = \alpha g_0 g_0^\top$ with $\alpha = (p-2)\|u\|^{p-4}$, $g_0 = J^\top u$, and metric $B = J^\top J + \varepsilon I$. Then $(\lambda B + \alpha g_0 g_0^\top)^{-1}$ admits the Sherman–Morrison closed form $\frac{1}{\lambda} B^{-1} - \frac{\alpha/\lambda^2}{1+(\alpha/\lambda)g_0^\top B^{-1}g_0} B^{-1} g_0 g_0^\top B^{-1}$, turning per-iteration cost from $O(m^3)$ Cholesky into $O(m^2)$ two-solve.

III. EXPERIMENTS

a) Setup: We benchmark 12 methods (Table I) across 25 configurations from six families: log-sum-exp on synthetic 1000×1000 data and on the LIBSVM datasets [15] a9a and mushrooms; nonlinear equations $\frac{1}{p}\|Ax - b\|^p$, $p \in \{2, 3, 4, 5, 6\}$; the Chebyshev chain of [1] with $n \in \{200, 500, 1000, 2000\}$; standard 2-D Rosenbrock [16]; and Rosenbrock-NLE, $\frac{1}{p}\|u\|^p$ for $p \in \{2, 3, 4, 5, 6, 8\}$. All methods stop at $f(x_k) - f^* < \varepsilon = 10^{-8}$ with $\gamma_0 = 1$ and identical adaptive-search parameters (Appendix A). The full suite runs in ≈ 25 min on a single CPU core.

b) Headline result: iteration speed-up: Figure 1 shows the iteration count to reach $\varepsilon=10^{-8}$ for `gns_inexact` versus `agns_lookahead` across the Rosenbrock-NLE p -sweep and the Chebyshev dimension sweep. The acceleration is consistent and substantial: a **1.6–2.3 $\times$** reduction on Rosenbrock-NLE (largest at $p = 2$, smallest at $p = 8$) and a **1.30–1.42 $\times$** reduction on Chebyshev. The comparison is apples-to-apples: both methods use the same Fisher or $J^\top J$ +Fisher approximation and the same γ_0 .

c) Convergence vs. all baselines: Figure 3 (appendix) places AGNS in the context of every relevant Newton-type baseline on a representative non-convex problem from each family. Three observations stand out: (i) on Rosenbrock-NLE ($p = 5$), every GNS-style method reaches $\varepsilon = 10^{-8}$ within 60 iterations whereas plain gradient descent takes ≈ 2000 , with AGNS-Lookahead the fastest among Newton-type methods; (ii) on the Chebyshev chain the gap between AGNS-Lookahead and unaccelerated GNS variants is narrower in iteration count but still visible; (iii) the iterate-anchored AGNS ablation tracks the lookahead variant but is consistently slightly slower, motivating the lookahead default.

d) Wall-clock impact of WSM: Figure 2 shows $f - f^*$ versus wall-clock time on $n = 100, m = 200, p = 4$ nonlinear equations. The exact-Cholesky baseline reaches ε in 0.110 s; `gns_wsm` and `agns_wsm_lookahead` reach it in 0.003 s, a **30–40 $\times$** speedup with no loss of accuracy. The gap grows cubically in m . Matrix-solve count (Fig. 6, appendix) shows that AGNS-WSM also halves the number of solves needed.

e) Where does AGNS not help?: We deliberately report negative findings. On log-sum-exp (a9a, $\mu=0.1$; mushrooms), GNS-Exact attains quadratic local convergence in 18 and 24 iterations respectively, fewer than the 40 and 57 taken by AGNS-Lookahead (Fig. 5, appendix). The Hessian is well-conditioned near the optimum so unaccelerated Newton already enjoys quadratic convergence; AGNS adds momentum overhead without altering the asymptotic regime. AGNS is not a free lunch: it accelerates methods bottlenecked by gradient

regularisation, not those already in the quadratic-convergence regime.

f) Robustness: On Rosenbrock-NLE $p \in \{6, 8\}$ the gradient and FISTA baselines diverge cleanly: $\|\nabla f(x_0)\|_* \approx 6 \times 10^7$ at $p = 8$ from $x_0 = (-2, 2)$, and no power-of-two step yields a finite probe value. We report this as `line_search_diverged` on iteration 0. All five Newton-type methods, by contrast, converge cleanly because their step is bounded by $\|x_{k+1} - x_k\| \leq \gamma_k$. This dichotomy is a useful illustration of why gradient regularisation is preferable to Lipschitz-based step-size selection on polynomially-flavoured objectives.

IV. DISCUSSION

a) Limitations: AGNS is a heuristic. The Lemma-1 condition holds at y_k for the lookahead variant, but combined with momentum it does not directly imply monotone descent in $f(x_k)$. Gradient restart prevents pathological cases empirically but is not part of any rigorous bound. Producing an actual convergence theorem for AGNS—ideally matching the Monteiro–Svaiter rates of [10] for cubic Newton—is the natural next step.

b) Reproducibility: The full benchmark suite is reproducible by running `bash scripts/run_all.sh` in the accompanying codebase (Appendix A). All hyperparameters live in YAML configurations; each method run produces a `summary.json` and a per-method history pickle. Total: 157 runs.

c) Outlook: Subspace versions of AGNS along the lines of [10] and stochastic AGNS for empirical-risk minimisation [18] are natural extensions.

REFERENCES

- [1] A. Semenov, M. Jaggi, and N. Doikov, “Gradient-normalized smoothness for optimization with approximate Hessians,” *arXiv preprint arXiv:2506.13710*, 2025.
- [2] C. Cartis, N. I. Gould, and P. L. Toint, “Adaptive cubic regularisation methods for unconstrained optimization. Part I: motivation, convergence and numerical results,” *Mathematical Programming*, vol. 127, no. 2, pp. 245–295, 2011.
- [3] Y. Nesterov and B. T. Polyak, “Cubic regularization of Newton method and its global performance,” *Mathematical Programming*, vol. 108, no. 1, pp. 177–205, 2006.
- [4] R. A. Polyak, “Regularized Newton method for unconstrained convex optimization,” *Mathematical Programming*, vol. 120, pp. 125–145, 2009.
- [5] K. Mishchenko, “Regularized Newton method with global $O(1/k^2)$ convergence,” *SIAM Journal on Optimization*, vol. 33, no. 3, pp. 1440–1462, 2023.
- [6] N. Doikov, “Minimizing quasi-self-concordant functions by gradient regularization of Newton method,” *arXiv preprint arXiv:2308.14742*, 2023.
- [7] N. Doikov, K. Mishchenko, and Y. Nesterov, “Super-universal regularized Newton method,” *SIAM Journal on Optimization*, vol. 34, no. 1, pp. 27–56, 2024.
- [8] Y. Nesterov, “A method for solving the convex programming problem with convergence rate $O(1/k^2)$,” *Doklady Akademii Nauk SSSR*, vol. 269, no. 3, pp. 543–547, 1983.
- [9] A. Beck and M. Teboulle, “A fast iterative shrinkage-thresholding algorithm for linear inverse problems,” *SIAM Journal on Imaging Sciences*, vol. 2, no. 1, pp. 183–202, 2009.
- [10] Y. Carmon, D. Hausler, A. Jambulapati, Y. Jin, and A. Sidford, “Optimal and adaptive Monteiro–Svaiter acceleration,” in *Advances in Neural Information Processing Systems*, vol. 35, pp. 20338–20350, 2022.

TABLE I: Methods compared. *Apx. Hess.*: uses a PSD approximation of $\nabla^2 f$ instead of the true Hessian.

Registry key	Algorithm	Reference	Apx. Hess.
gns_exact	GNS, $H = \nabla^2 f$	[1]	no
gns_inexact	GNS with Fisher / Weighted Gauss–Newton approx.	[1]	yes
gns_wsm	GNS with rank-1 Fisher and WSM closed-form solve	this work	yes
agns_lookahead	AGNS-Lookahead (ours) — Hessian at y_k	this work	yes
agns_iterate	AGNS with iterate anchor (ablation)	this work	yes
agns_wsm_*	AGNS with WSM solver (lookahead, iterate)	this work	yes
super_newton	Super-Universal Newton method	[7]	no
cubic_newton	Adaptive Cubic Regularisation	[3], [2]	no
gradient	Gradient method, backtracking line search	[17]	—
fast_gradient	Nesterov accelerated gradient method (FISTA-style)	[9]	—

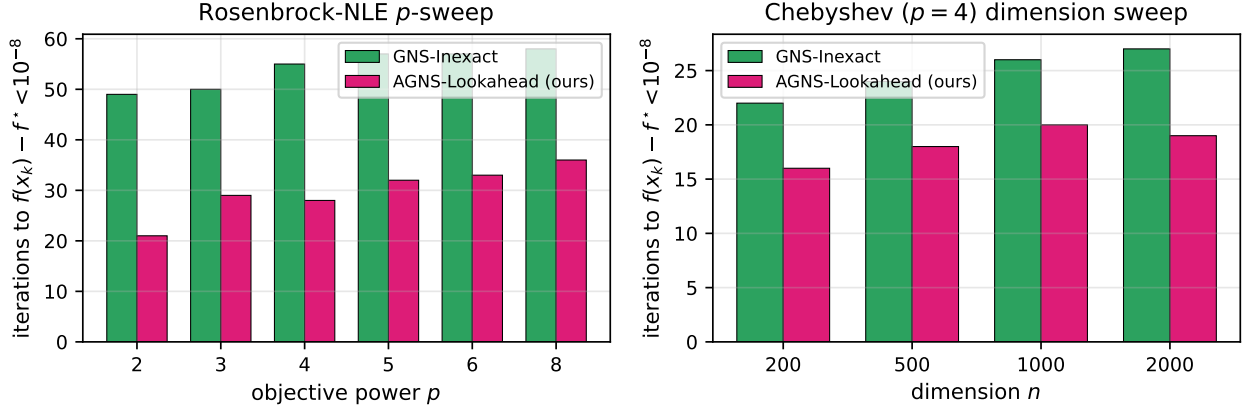


Fig. 1: **Iteration speed-up of AGNS-Lookahead over GNS-Inexact.** *Left:* Rosenbrock-NLE for $p \in \{2, 3, 4, 5, 6, 8\}$, $x_0 = (-2, 2)$, $f^* = 0$. *Right:* Chebyshev polynomial chain at $p = 4$, $n \in \{200, 500, 1000, 2000\}$. The y -axis is the number of outer iterations to reach $f(x_k) - f^* < 10^{-8}$. Both methods use identical Hessian approximations, identical $\gamma_0 = 1$, and the same adaptive-search rule. AGNS reduces iteration count by 1.6 – $2.3\times$ on Rosenbrock-NLE and 1.30 – $1.42\times$ on Chebyshev. Per-iteration convergence curves at every p are in Fig. 4 (appendix).

- [11] B. O’Donoghue and E. Candès, “Adaptive restart for accelerated gradient schemes,” *Foundations of Computational Mathematics*, vol. 15, no. 3, pp. 715–732, 2015.
- [12] M. A. Woodbury, “Inverting modified matrices,” *Statistical Research Group, Princeton University, Memorandum Report 42*, 1950.
- [13] Y. Nesterov, “Modified Gauss–Newton scheme with worst case guarantees for global performance,” *Optimization Methods and Software*, vol. 22, no. 3, pp. 469–483, 2007.
- [14] F. Kunstner, P. Hennig, and L. Balles, “Limitations of the empirical Fisher approximation for natural gradient descent,” *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [15] C.-C. Chang and C.-J. Lin, “LIBSVM: A library for support vector machines,” *ACM Transactions on Intelligent Systems and Technology*, vol. 2, no. 3, pp. 1–27, 2011.
- [16] H. H. Rosenbrock, “An automatic method for finding the greatest or least value of a function,” *The Computer Journal*, vol. 3, no. 3, pp. 175–184, 1960.
- [17] Y. Nesterov, *Lectures on Convex Optimization*, vol. 137. Springer, 2018.
- [18] J. Martens, “New insights and perspectives on the natural gradient method,” *Journal of Machine Learning Research*, vol. 21, no. 146, pp. 1–76, 2020.

APPENDIX A

HYPERPARAMETERS AND REPRODUCIBILITY DETAILS

All experiments use $\gamma_0 = 1$ for the second-order step-size, $\varepsilon = 10^{-8}$ for the stopping criterion, and the adaptive search of [1], eq. (16): on acceptance, γ_k doubles; on rejection, γ_k halves

with floor $\gamma_{\min} = 10^{-9}$. The inner adaptive-search loop is capped at 40 probes (the cap is never reached on any successful run). Gradient methods use $L_0 = 1$, $L_{\min} = 10^{-5}$, and a Lipschitz backtracking line search with cap of 30 probes; we additionally reject any trial point with non-finite f value (the original code accepted `nan` silently).

The Hessian approximations used are: *Weighted Gauss–Newton* for log-sum-exp, $H(x) = \frac{1}{\mu} A^\top \text{diag}(\text{softmax}(Ax)) A$; the *Fisher term* for nonlinear equations, $H(x) = (p - 2) \|u(x)\|^{p-4} J^\top u (J^\top u)^\top$; and $J^\top J$ + Fisher for the Chebyshev chain (the $J^\top J$ term is dense and x -dependent, so the rank-1 WSM solver does not apply there).

The complete pipeline is invoked via

```
bash scripts/download_data.sh
bash scripts/run_all.sh
python tests/test_oracles.py
python tests/test_wsm_identity.py
python tests/test_methods_smoke.py
```

producing 25 result directories, 31 PNG plots, 182 history pickles, 29 `summary.json` files, and zero NaN/Inf values

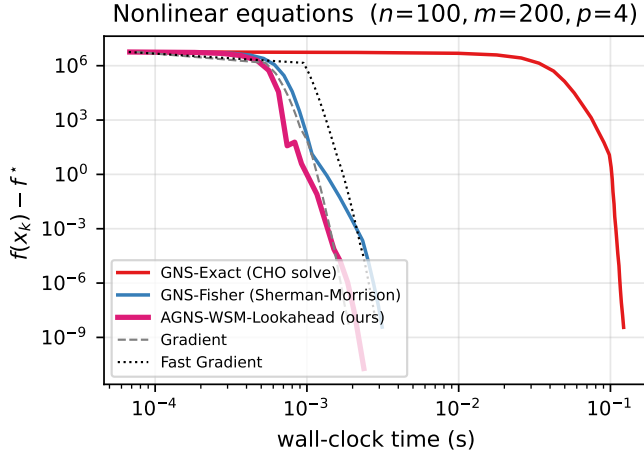


Fig. 2: **Wall-clock comparison** on $f(x) = \frac{1}{4} \|Ax - b\|^4$, $A \in \mathbb{R}^{100 \times 200}$, with metric $B = A^\top A + 10^{-8}I$. The WSM-based solvers reach the target two orders of magnitude faster than GNS-Exact. Both axes are logarithmic. Matrix-solve count for the same run is in Fig. 6.

across 157 method runs. All paper figures can be regenerated from those results by running the accompanying notebook `notebooks/make_paper_figures.ipynb`. The repository also ships an extended audit (finite-difference checks for every oracle gradient and Hessian, WSM identity check) that completes in < 1 s.

APPENDIX B ADDITIONAL FIGURES AND TABLES

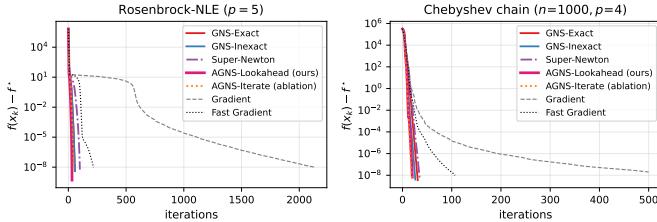


Fig. 3: **Convergence on representative non-convex benchmarks.** *Left:* Rosenbrock-NLE $p = 5$ from $x_0 = (-2, 2)$. *Right:* Chebyshev polynomial chain, $n = 1000, p = 4$, $x_0 \sim U[0, 1]^n$. Trajectories truncated at the first iterate satisfying $f(x_k) - f^* < 10^{-8}$; logarithmic y -axis. AGNS-Lookahead (magenta, ours) reaches the target before any GNS variant on both problems; AGNS-Iterate (orange dotted) tracks but never beats AGNS-Lookahead.

APPENDIX C ALGORITHM DETAILS

The full AGNS implementation in `src/methods.py` adds two guards beyond Algorithm 1: (i) if the inner adaptive search exhausts its budget (40 probes) without an accepted step, the trial point is set to y_k as a safe fallback rather than

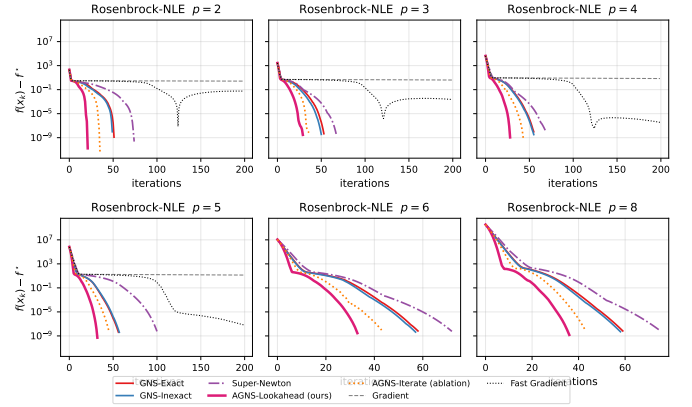


Fig. 4: **Per-iteration convergence on the Rosenbrock-NLE p -sweep.** Six panels for $p \in \{2, 3, 4, 5, 6, 8\}$, all from $x_0 = (-2, 2)$, $f^* = 0$, $\varepsilon = 10^{-8}$. AGNS-Lookahead (magenta) is the fastest method on every panel. Plain gradient and FISTA (gray, black) diverge cleanly at $p \geq 6$. The horizontal axis is truncated at 200 iterations for readability; tails of *Gradient* on $p \in \{2, 3, 4, 5\}$ extend to several thousand iterations (Table II).

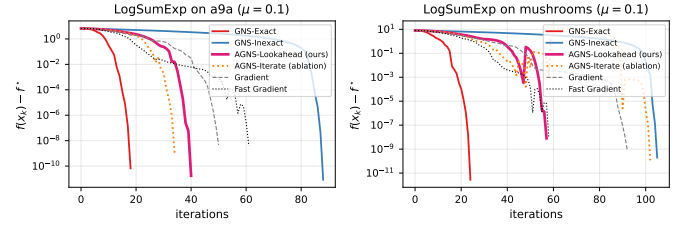


Fig. 5: **LogSumExp on real LIBSVM data** (a9a, mushrooms; $\mu = 0.1$). The benchmark where AGNS does *not* accelerate convergence: GNS-Exact already attains quadratic local convergence and reaches the target in 18 and 24 iterations respectively, fewer than AGNS-Lookahead’s 40 and 57. Note that the residuals shown are not zero; the true optimum is offset by the construction of [1].

committed unconditionally; and (ii) if $\|\nabla f(y_k)\|_* < 10^{-8}$, the Newton step is skipped (we are already at a stationary point near y_k).

For the rank-1 WSM solver, when $p = 2$ the Fisher coefficient $\alpha = (p-2) \|u\|^{p-4}$ vanishes and the formula degenerates to $\Delta = -(\lambda B)^{-1} \nabla f$, i.e. a preconditioned gradient step (the Hessian-approximation contribution is identically zero for quadratic f , which is consistent).

For GNS-Exact on indefinite Hessians (e.g. Chebyshev chain far from the optimum), Cholesky factorisation initially fails; we catch `LinAlgError`, halve γ_k to tighten the regularisation, and retry within the same inner loop. This is mathematically equivalent to a rejected step.

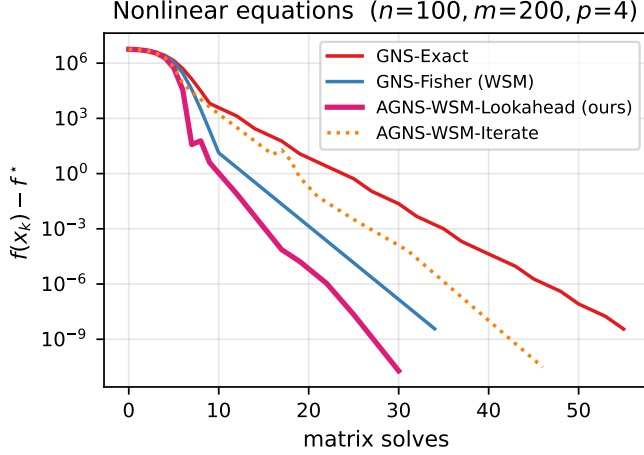


Fig. 6: **Matrix-solve count vs. accuracy** on nonlinear equations ($n = 100, m = 200, p = 4$). Each iteration of every Newton-type method costs one linear solve (GNS-Exact: dense Cholesky; GNS-WSM and AGNS-WSM: Sherman–Morrison closed form). AGNS-WSM-Lookahead needs roughly half as many solves as GNS-Exact to reach the same residual.

TABLE II: Iterations to $\varepsilon = 10^{-8}$ on the Rosenbrock-NLE p -sweep ($x_0 = (-2, 2)$). Apples-to-apples: every method uses the same Fisher / $J^\top J$ +Fisher approximation. “—” marks line_search_diverged. Best Newton-type method per row in **bold**.

p	GNS-Ex.	GNS-Inex.	Sup.-N.	AGNS-Look.	Grad.	FastG.
2	51	49	74	21	6852	407
3	53	50	67	29	4331	265
4	55	55	69	28	2892	238
5	56	57	100	32	2132	218
6	58	57	72	33	—	—
8	59	58	75	36	—	—

TABLE III: Iterations and wall-clock time to $\varepsilon = 10^{-8}$ on the Chebyshev chain dimension sweep ($p = 4$). Wall-clock is dominated by the $O(n^3)$ Cholesky of GNS variants.

n	GNS-Inexact	AGNS-Lookahead	Speedup
200	22 0.073 s	16 0.057 s	1.38× iter, 1.28× time
500	24 1.32 s	18 1.12 s	1.33× iter, 1.18× time
1000	26 8.11 s	20 6.46 s	1.30× iter, 1.26× time
2000	27 11.97 s	19 8.26 s	1.42× iter, 1.45× time